

ADR-004 : Migration from JavaScript to TypeScript

Contexte

As the project's codebase grows and becomes more complex (especially with our front-end and back-end applications), JavaScript's dynamic typing shows its limits. Runtime type errors are becoming more frequent, and large-scale refactoring is risky. In order to improve maintainability, code reliability, and the developer experience (autocompletion, integrated documentation), an assessment of the typing system was conducted.

Options considérées

- Stay in JavaScript : No migration cost or additional compilation step, but typing-related errors continue to appear only at runtime.
- Migrate to TypeScript : Adds strong static typing, detects errors at compile time, and represents the current standard for robust applications.

Décision

Migrate the codebase to TypeScript.

Justification : The initial investment to configure TypeScript and type the existing code is largely offset by the reduction of bugs in production and the safety provided during modifications. Furthermore, TypeScript integrates natively and optimally with the core technologies of our architecture, making it a natural and long-term choice for the project.

Conséquences

- Positive : Error detection during development rather than at runtime.
- Positive : Excellent autocompletion in editors and self-documented code.
- Positive : Much safer and faster refactoring.
- Négative : Adds a compilation (transpilation) step to the build process.